NET ZERO
TRANSITION PLAN

ESG Discovery • Carbon Accounting • Financial Modeling • Reporting Alignment

ISSB / IFRS S1 & S2	ESRS	GRI	GHG Protocol	Paris Agreement

45,805 tCO_2e	18,000 tCO_2e	USD 1.5M	81.3%
Baseline Emissions (2023)	2030 Target	Total CAPEX	Annual ROI

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Project Details
Q4 2024 — Q1 2025
Tools: SQLite • Power BI • DAX • SQL • Excel (Advanced)
Deliverables: Database • Dashboard • Financial Model • Report

1. Project Overview & Defined Scope

This project was developed as a comprehensive ESG engagement simulation covering the full lifecycle of a Net Zero Transition Plan: from initial client discovery and maturity assessment, through data collection and carbon accounting, to financial modeling and framework-aligned reporting. It demonstrates end-to-end ESG project execution capability across discovery, data, and reporting workstreams.

Project Scope

- Carbon accounting — Scope 1, 2 & 3
- Abatement strategy & financial modeling
- Multi-scenario emissions trajectory analysis
- Implementation roadmap (2024–2040)
- Risk assessment with mitigation strategies

Key Deliverables

- 14-table normalized SQLite database
- 6-page Power BI dashboard (30+ visuals)
- Excel-based ESG data models & templates
- Full ROI & scenario financial model
- Audit-ready documentation & traceability log

2. ESG Discovery & Client Maturity Assessment

Before any data collection or reporting work began, a structured ESG discovery process was conducted to establish the client's current maturity level, identify gaps, and define realistic project scope and timelines.

Discovery Process

1. Stakeholder mapping — identified responsible parties across finance, operations, procurement, and leadership for each data domain.
2. Business operations translation — mapped operational activities to GHG Protocol emission categories (Scope 1, 2, 3) and relevant ISSB/ESRS disclosure topics.
3. Data availability audit — assessed which emissions data existed in structured form, which required estimation, and which required new data collection processes.
4. Maturity scoring — evaluated the client across five ESG dimensions using a 4-level scale (see table below).
5. Scope definition — documented project boundaries, included/excluded emission categories, and agreed timelines with clear deliverable milestones.

Client ESG Maturity Assessment

ESG Dimension	Maturity Level	Current State	Gap Identified
Carbon Accounting	Level 2 — Developing	Scope 1 & 2 tracked; Scope 3 absent	Scope 3 data collection framework required
ESG Reporting	Level 1 — Initial	No formal ESG report produced	Full ISSB/ESRS alignment needed
Governance	Level 2 — Developing	No board-level ESG committee	Governance structure to be formalized
Data Infrastructure	Level 2 — Developing	Fragmented; no central ESG data model	Centralized database + templates required
Supplier Engagement	Level 1 — Initial	No supplier ESG requirements in place	Supplier engagement program (M4) needed

Maturity levels: Level 1 (Initial) — ad hoc, no formal processes. Level 2 (Developing) — partial processes, inconsistent. Level 3 (Defined) — documented, consistent. Level 4 (Optimized) — measured, continuously improved.

Discovery Outcome: Defined Project Scope

Priority 1: Build Scope 3 data collection framework and complete full GHG inventory
Priority 2: Develop financial model to support board-level Net Zero decision
Priority 3: Align outputs to ISSB S1/S2 and GRI disclosure requirements
Priority 4: Produce audit-ready documentation with full data traceability
Out of scope (Phase 1): BRSR, external assurance, supplier audits — recommended for Phase 2

3. ESG Frameworks Applied

This project aligns outputs to four internationally recognized frameworks. The framework mapping ensures that data collected and reported in this engagement can be directly used in formal ESG disclosures.

Framework	Role in This Project	Key Requirements Addressed
ISSB / IFRS S2	Primary climate disclosure standard	Governance, Strategy, Risk Management, Metrics & Targets (TCFD-aligned)
ISSB / IFRS S1	Sustainability materiality umbrella	Financial materiality assessment; disclosure alongside financial statements
GRI (Universal Standards)	Stakeholder-facing reporting	GRI 305 (Emissions) — Scope 1, 2, 3 disclosures; GRI 3 (Materiality)
GHG Protocol	Carbon accounting methodology	Scope 1/2/3 boundary setting, emission factors, calculation methodology
Paris Agreement	Net Zero target definition	1.5°C pathway alignment; carbon offset validity for 2030 Net Zero claim

ISSB S1/S2 Alignment Map

The four pillars of ISSB S2 (derived from TCFD) were directly addressed in this engagement:

ISSB S2 Pillar	This Project's Coverage	Deliverable
Governance	Identified absence of board ESG committee; documented gap and recommended structure	Maturity Assessment (Section 2)
Strategy	Scenario analysis (BAU vs Net Zero); financial impact on business model through 2040	Scenario Model (Section 5)
Risk Management	5 climate-related risks identified, severity scored, mitigation strategies defined	Risk Register (Section 8)
Metrics & Targets	Full Scope 1/2/3 inventory; 2030 & 2040 emission targets with annual milestones	GHG Inventory (Section 4)

4. Data Architecture & Collection Framework

A critical component of this engagement was designing a data infrastructure capable of supporting audit-ready ESG reporting. The architecture combines a relational database backend with Excel-based data entry templates accessible to client teams without technical expertise.

Database Architecture

A normalized SQLite star schema was designed with 14 tables across two layers:

Layer	Tables	Purpose
Core (10 tables)	emissions_baseline, scope1_activities, scope2_consumption, scope3_categories, abatement_measures, financial_projections, carbon_pricing, scenarios, milestones, risk_register	Primary data storage; each table maps to a specific GHG Protocol category or financial module
Analytics (4 tables)	kpi_dashboard, scenario_comparison, measure_roi, progress_tracking	Pre-aggregated views feeding Power BI dashboard; recalculate on data refresh

Excel Data Collection Templates

To enable non-technical client teams to contribute data, four Excel templates were designed with built-in validation, dropdown controls, and error-checking:

Template	Data Collected	Validation Rules
Scope 1 Activity Log	Fuel type, quantity (L), vehicle/equipment ID, period	Emission factors auto-applied; outlier flags for >2σ deviations
Scope 2 Consumption	Meter readings, facility, grid zone, renewable certificates	Grid emission factors by region; RECs validated against certificates
Scope 3 Spend Data	Supplier name, spend category, NACE code, annual spend (USD)	EEIO emission factors applied; flags for high-spend unclassified suppliers
Abatement Tracking	Measure ID, deployment date, actual vs. projected reduction	Variance analysis auto-calculated; RAG status assigned

Data Validation & Audit Trail

All data inputs were subject to a three-step validation process to ensure audit readiness:

6. Completeness check — all required fields populated; missing data flagged with source and reason documented.
7. Reasonableness check — values compared against sector benchmarks (IPCC, IEA) and prior period data; anomalies investigated and resolved or explained.
8. Traceability log — each data point linked to source document (invoice, meter reading, supplier certificate) with date, responsible party, and version history.

Audit-Ready Documentation Standard

Every emission figure traceable to source document with unique reference ID
Calculation methodology documented per GHG Protocol requirements
Change log maintained for all data revisions with author and date
Boundary decisions (included/excluded activities) formally documented
Emission factors sourced from IPCC AR6, IEA 2023, and national grid data — version controlled

5. GHG Inventory — Carbon Accounting Results

The 2023 baseline inventory was constructed using the GHG Protocol Corporate Standard, operational control consolidation approach, with 100% coverage of Scope 1 and 2, and Scope 3 categories 1, 4, and 6 identified as material.

3,501 tCO_2e	3,444 tCO_2e	38,860 tCO_2e
Scope 1 — Direct	Scope 2 — Electricity	Scope 3 — Supply Chain
8% of total	8% of total	85% of total

Scope	Category	Activity Data Source	Method	Emissions (tCO_2e)
Scope 1	Stationary combustion	Fuel purchase invoices	IPCC Tier 2	2,150
Scope 1	Mobile combustion (fleet)	Fuel logs & odometer	IPCC Tier 2	1,351
Scope 2	Purchased electricity	Utility bills (12 months)	Location-based (IEA grid factor)	3,444
Scope 3 — Cat. 1	Purchased goods & services	Spend-based (EEIO)	GHG Protocol Scope 3 Standard	37,800
Scope 3 — Cat. 4	Upstream transport	Freight invoices + distance	Spend-based	780
Scope 3 — Cat. 6	Business travel	HR travel expense system	Distance-based	180
TOTAL	—	—	—	45,805

Note on Scope 3 completeness: Categories 2, 3, 5, 7–15 were screened and determined not material for this company's operational profile. Screening methodology and exclusion rationale documented in data appendix.

6. Abatement Strategy & Measure Prioritization

Five abatement measures were identified and evaluated using a Marginal Abatement Cost Curve (MACC) methodology, allowing direct comparison of measures across impact, cost, and timing dimensions.

ID	Measure	Scope	Abatement (tCO_2e)	CAPEX (USD)	ROI	Payback	Priority
M4	Supplier Engagement Program	S3 Cat.1	5,670	$200K	126.8%	0.44 yr	⭐ Critical
M1	Renewable Electricity (PPA)	Scope 2	3,444	$300K	-8.2%	1.09 yr	High
M3	Energy Efficiency Program	Scope 1	700	$200K	-72%	3.57 yr	Medium
M2	Fleet Electrification	Scope 1	675	$600K	-91%	11.1 yr	Low
M5	Logistics Optimization	S3 Cat.4	195	$200K	-92.2%	12.8 yr	Low
TOTAL	Combined Portfolio	—	10,684	$1.5M	81.3%	0.55 yr	—

M4 is the strategic lever — 53% of total abatement at the highest ROI

M4 alone generates more abatement than all other measures combined.
Despite negative individual ROIs, M1–M3 and M5 are justified for risk diversification and regulatory Scope 1/2 compliance.
Combined portfolio payback of 6.6 months makes the full $1.5M investment financially compelling.
M4 requires 40% supplier participation by 2027 — negotiations must start Q4 2024 (3-year lead time).

7. Scenario Analysis

Two scenarios were modeled to quantify the financial consequences of action vs. inaction, framing Net Zero as a financial decision rather than purely an environmental one. Scenario analysis is a core requirement under ISSB S2 and TCFD.

Scenario	2030 Emissions	Carbon Cost @ $80/tCO₂e	vs. Net Zero Pathway
Business as Usual (BAU)	52,000 tCO_2e	USD 4.16M/yr	+USD 2.72M more annually
Net Zero Pathway	18,000 tCO_2e	USD 1.44M/yr	Baseline
Cost avoidance	34,000 tCO_2e reduction	USD 2.72M/yr	—

Without intervention, emissions grow 13% from 2023 baseline to 52,000 tCO_2e by 2030. Full Net Zero implementation achieves a 61% reduction vs. the BAU trajectory.

Carbon Price Sensitivity

@ $80/$tCO_2e$ → ROI: 81.3% (base case)
@ $50/$tCO_2e$ → ROI: 13% (still positive)
@ $30/$tCO_2e$ → ROI: negative (M1 PPAs protect)

2030 vs. 2040 Pathway

2030: Net Zero via offsets (18,000 tCO_2e + 18K offsets)
2040: True Net Zero — 0 emissions, no offsets needed
Both pathways Paris Agreement compliant

8. Climate Risk Register

Climate-related risks were identified and assessed in alignment with ISSB S2 and TCFD risk classification (transition risks and physical risks). Each risk includes probability, impact, and mitigation strategy.

Risk ID	Type	Risk	Severity	Financial Impact	Mitigation
R1	Execution	M4 Supplier Participation fails (<40%)	CRITICAL	Loss of $453.6K annual benefit	Start Q4 2024; phased rollout; incentives
R2	Transition	Carbon price lower than $80/t$CO_2$e assumption	MEDIUM	ROI drops 81% → 13%	Lock M1 PPAs; monitor Chile/Peru carbon policy
R3	Execution	EV charging infrastructure unavailable for M2	MEDIUM	M2 delayed to 2028+	Partner with charging networks; pilot program
R4	Physical	Technology underperformance vs. guarantees (M1, M3)	LOW	Reduced abatement, extended payback	Proven technology only; performance guarantees
R5	Economic	Economic downturn — $1.5M CAPEX unaffordable	MEDIUM	Project delays; missed 2030 target	Phased investment; high-ROI first; green financing

9. Financial Model & ROI Analysis

The financial model was built in Excel with linked tables feeding into Power BI. It calculates project-level and portfolio-level ROI, payback periods, and 5-year cumulative benefits, with a sensitivity analysis module for carbon price assumptions.

$1.5M	$2.72M	6.6 months	$10.25M
CAPEX Investment	Annual Cost Avoided	Payback Period	5-Year Net Benefit

Financial Metric	Value	Notes
Total CAPEX Required	USD 1,500,000	Phased 2024–2027
Annual Carbon Cost Savings (2030)	USD 2,720,000	At $80/t$CO_2$e
Annual ROI (portfolio)	81.3%	On $1.5M investment
Payback Period	0.55 years (6.6 months)	Portfolio level
5-Year Net Benefit (2025–2030)	USD 10,250,000	Cumulative
Annual Carbon Cost — BAU (2030)	USD 4,160,000	Without intervention
Annual Carbon Cost — Net Zero (2030)	USD 1,440,000	Including offset purchases
Annual Cost Avoidance	USD 2,720,000	BAU vs. Net Zero

10. Implementation Roadmap

The roadmap is structured in six phases with clear milestones, CAPEX schedule, and cumulative abatement targets. It is designed to be used as the primary project management reference throughout execution.

Phase	Period	Actions	Cumulative Abatement	CAPEX
Initiation	Q4 2024	Board approval; budget secured; M4 negotiations begin; M1 PPAs locked	—	—
Quick Wins	2025	M1 renewable electricity deployed; M3 energy efficiency launched	4,144 tCO_2e (9%)	$500K
Scaling	2026	M2 fleet electrification begins; M5 logistics optimization launched	5,014 tCO_2e (11%)	$800K
Critical Deployment	2027	M4 supplier engagement goes live at scale (40% participation)	10,684 tCO_2e (23%)	$200K
Gap Closure & Optimization	2028–2030	Optimize all measures; close 69% abatement gap via offsets/redesign	18,000 tCO_2e	—
Full Decarbonization	2031–2040	Additional measures + technology improvements; eliminate offset dependency	~0 tCO_2e	TBD

11. Recommendation & Next Steps

PROCEED IMMEDIATELY — Financial & Strategic Case is Compelling

81.3% annual ROI on $1.5M investment with 6.6-month payback justifies immediate action.
Net Zero 2030 via carbon offsets is Paris Agreement compliant and financially attractive.
2040 pathway enables full decarbonization without offset dependency.

Immediate next steps (Q4 2024):

1. Board approval and $1.5M budget allocation
2. Start M4 supplier engagement negotiations — 3-year lead time is non-negotiable
3. Lock renewable energy PPAs for M1 before year-end
4. Activate monthly GHG tracking dashboard
5. Assign cross-functional steering committee (finance, ops, procurement, legal)

Phase 2 recommendations (2025):

- External assurance of GHG inventory (ISAE 3410)
- Supplier ESG questionnaire rollout
- BRSR alignment assessment (if targeting Indian capital markets)
- Full ESRS gap analysis for EU regulatory exposure

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SQLite • Power BI • Excel (Advanced) • DAX • SQL • GHG Protocol • ISSB • GRI